# COMMERCIAL ITEM DESCRIPTION

## AUDIOMETER, AUTOMATIC, PORTABLE

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers an automatic Bekesy-type audiometer with recording mechanism and carrying case.

Salient characteristics.

The audiometer shall comply with all applicable requirements of ANSI S3.6, Specifications for Audiometers and UL544, Electric Medical and Dental Equipment. In addition, the unit shall comply with the requirements of this specification. In the event of conflict, the requirements of this specification shall take precedence.

The audiometer shall be Bekesy type with six (6) discrete frequencies. The components of the audiometer shall include a recording mechanism, a patient handswitch, and earphone assembly, audiogram cards, and a portable carrying case.

Attenuation. The audiometer shall possess an attenuator, which shall be controlled by a patient handswitch. The attenuator shall have an attenuation range of 100 decibels (dB). The attenuator shall have a linear non-step structure. The maximum tolerance shall be ± 1 dB for any 10 dB increment, and shall not exceed ± 3.0 dB cumulative for the entire attenuation range or any portion of the range. The system shall be calibrated such that the total dynamic range of attenuation will correspond to a Hearing Threshold Level Range on the audiogram card of 0 dB to 100 dB.


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Frequency. Test frequencies shall be fixed in discrete steps of 500, 1000, 2000, 3000, 4000, and 6000 Hz. Frequency error shall not be greater than ± 3% for any given frequency. Total Harmonic Distortion shall be a least -35 dB in reference to the desired test frequency.

Earphones. Earphones shall be of the dynamic moving coil type. The left and right earphones shall be marked with different colors for identification. Earphones shall be calibrated at each of the six (6) discrete frequencies. Calibration shall be within ± 3 dB for all frequencies with exception 6000 Hz which shall be within ± 5 dB. The sound pressure level of the fundamental frequency shall be at least 35 dB above that of any harmonic of that fundamental. Any signal from the earphone not being used for test purposes will have a sound pressure level at least 10 dB below the Hearing Threshold Reference Level, except it need not be more than 70 dB below the signal from the active earphone. The headband shall be constructed of sufficient strength to hold the earphones securely in place.

Controls. All controls shall be located on the front panel of the unit. The audiometer shall be provided with following control functions:

Tests. This control shall either start the audiometric test or stop the audiometric test. A lighted display shall indicate when this function is activated. The test shall be capable of being stopped at any time. Activation of the test function shall cause the audiogram marking pen to drop, audiogram card to advance, and attenuation decrease (increase in sound pressure level for test frequency). The length of the test will be approximately six (6) minutes. During this interval each ear (left then right) will be tested at each of the six Hz. Each test tone will be applied to an earphone. At the end of each test tone, the tone shall be interrupted which will indicate the subject has acquired threshold. The indication of this interruption period at the end of each test tone shall be characterized by a larger excursion at the frequency change line on the subject is responding correctly. During the test off, the handswitch shall be disabled and the attenuation level returned to 0 dB reference. Automatic test stop shall be accomplished when the audiogram card has advanced its entire length.



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-10 dB. When activated, the control shall lower the test tone sound pressure level at the earphone by 10 ± 1 dB and stop advancement of the audiogram card. An event marker shall be recorded on the audiogram card by a vertical line during the interval of "-10 dB".

Pulsed. When activated, this control shall cause the continuous test tone to pulse. A visual indication light shall be on during the "pulsed" state.

Mid-Delay. This control shall cause the audiogram card to stop and manually override the handswitch causing attenuation toward 0 reference when the test tone is completed at 6000 Hz left ear. The delay interval shall be approximately 15 seconds. A visual indication light shall be on when this control is activated.

Trial. This control shall be provided for the application of group trial.

Power. A power control shall be provided to enable/disable the unit. When activated, an indication light shall be illuminated.

Subject Control Handswitch. Each audiometer shall be provided with a control switch, assembled in an insulated grip with at least three (3) feet of conductor length. The handswitch shall control the signal intensity across the hearing threshold range so that the test tone sound pressure level will decrease when the button is depressed and increase when the button is released.

Accessories.

Pen. Each audiometer shall be supplied with four (4) pens: two (2) red an two (2) black. Each pen shall provide recording ink for at least 500 audiogram cards.

Carrying case The instrument shall be enclosed in a plastic carrying case. Suitable storage space shall be provided for all accessories. The audiometer with carrying case shall not weigh more than 35 pounds and dimensions shall not be greater than 10 x 12-1/2 x 22-1/2 inches overall.

Audiogram Cards. One hundred audiogram cards suitable for recording results of examination shall be furnished with each audiometer. Audiogram cards provided shall be in accordance with DPSC drawing No. 24152.



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Workmanship. The audiometer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One audiometer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificate of Quality. When available, certificates of quality may be furnished in lieu of actual performance of such testing. The certificate shall be supplied by the manufacturer of the component or material listing the specified test method and test results obtained.



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Tests. Tests shall be conducted to determine compliance with the specified requirements. Testing shall include but not be limited to the following:

Accuracy of tone frequency. The accuracy of the tone frequencies generated by the audiometer shall be determined using a calibrated frequency counted. Frequency error shall not be greater than that specified.

Purity of tone. The sound pressure levels of the fundamental frequency an its harmonics shall be measured with the earphone coupled to a National Bureau of Standards Coupler 9-A. Measurements shall be made at all frequencies. The difference between the levels of the fundamental and harmonic shall be within specified requirements.

Accuracy of Attenuator. Measurement of attenuator accuracy shall be made electrically at the input to the earphone terminated in a dummy load. Accuracy shall be within specified tolerances.

Accuracy of Sound Pressure Levels. The accuracy of the sound pressure levels at each frequency shall be determined with the earphone coupled to a National Bureau of Standards Coupler 9-A. The accuracy of sound pressure levels at each frequency after a minimum of one hour of operation shall also be determined. The sound pressure levels obtained shall conform to specified requirements.

Sound from Test Earphones. The rms weighted sound pressure, due to all frequencies except the signal frequency and its harmonics, shall be determined with the earphone coupled to a level shall be within specified limits.

Sound from Second Earphone. The sound pressure level in the earphone not being used for test purposes shall be measured to determine compliance with the requirements. This test shall be performed by measuring the electrical signal across a dummy load which simulates the impedance of the earphone.



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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standard, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

| NSN | ITEM IDENTIFICATION |
|---|---|
| 6515-00-533-6644 | AUDIOMETER, AUTOMATIC, PORTABLE |

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.



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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4421



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